What Is a Home Worth? Large-Scale Ensemble Learning for U.S. Housing Prices

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GitHub Repository: https://github.com/SatrunsDream/DSC148-Project

Introduction

In today's competitive housing market, overpaying for a home is an all-too-common pitfall, and one that can lead to missed opportunities and costly investment mistakes. Accurately determining a home's expected price and value could be a crucial aspect in real estate, benefiting both sellers and buyers. This project aims to build a model that predicts home prices and assesses their value, helping to identify if a property's price is reasonable.

1 Datasets

1.1 Identify Our Datasets

The USA Real Estate dataset, available in three parts (first_half.csv, second_half.csv, and third_half.csv) originates from a Kaggle dataset[1]. This dataset provides detailed information on real estate listings across the United States, categorized by state and zip code. It consists of 2,226,382 entries, with each row representing an individual property listing, and 12 columns containing various attributes of the property, such as the number of bedrooms, bathrooms, and other relevant features.

Other datasets were also used to create context for the market of a property's area (we're basing it on the zip code). One dataset used was the ZIP Code Tabulation Areas Dataset (stored as national-zcta-data.csv)[2], which contains geographic identifier codes, names, area measurements, and representative latitude and longitude coordinates for zip codes based on the 2020 Census tabulation blocks.

Another one is the US Population Density by Zip Code, City, and State Dataset (stored as zip_population.xlsx)[3], which contains 2010 census data for every zip code in the United States, including the corresponding City, State, and latitude/longitude coordinates for the center of the zip code.

Another one used was the Offenses Known to Law Enforcement Dataset (stored as fbi-crime.csv)[4], which contains the volume of violent crime and property crime, as reported by 12 months of complete offense data for 2015, as reported by city and town law enforcement agencies that contributed data to the Uniform Crime Reporting Program.

Another dataset used was the Institutional Characteristics Dataset (stored as hd2023.csv)[5], which contains detailed information for every institution in the 2023-24 IPEDS universe, including name, address, city, state, zip code, various URL links (e.g., homepage, admissions, financial aid, net price

[1] USA Real Estate dataset
[2] ZIP Code Tabulation Areas
[3] US Population Density by Zip Code, City, and State
[4] Offenses Known to Law Enforcement
[5] Institutional Characteristics

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calculator), institutional characteristics (e.g., control, level, highest degree offered, Carnegie classifications), geographical location variables (e.g., county, congressional districts, statistical areas, urbanization degree), and identifies active institutions participating in Title IV federal financial aid programs.

The last dataset used was the Property Taxes by State Dataset (stored as propertytax-by-state.csv)[6], which provides state-level property tax information, including effective tax rates for owner-occupied housing in 2022 and 2023, as well as state rankings based on tax rates, offering insights into regional tax burdens.

1.2 Perform EDA

1.2.1 Data Cleaning

We first began cleaning by dropping rows with missing 'zip_code' because we felt that it couldn't be reliably imputed and where the 'state' wasn't a US state or DC, e.g., Guam or Puerto Rico. We also made sure every feature was of the correct type. We then began the imputing side of things after a good chunk of the original dataset had missing values. We filled in missing values using hierarchical imputation by first using the median zip code values, then median state values, and median global values as a final fallback. Specifically to handle the missing values of columns 'city' and 'house_size', we imputed first using zip code, however for city we then filled in any remaining missing values with state mode values, and dropping the remaining rows, and for 'house_size', we filled in any missing values with median grouped city, state values, and then filled in median state values as a final fallback. We created a binary feature 'first_sale', which was created for homes with missing 'prev_sold_date' values, marking homes that had never been sold before. Temporal data in 'prev_sold_date' was adjusted to account for future dates beyond 2024. Categorical features, such as 'brokered_by' and 'street', were cleaned by filling missing values with 'Not Specified'. Any duplicates were then dropped.

After cleaning up the original USA Real Estate dataset, we then moved on to merging the other datasets onto the original dataset. We merged the dataframe 'zipcode_land' (ZIP Code Tabulation Areas Dataset) onto 'realtor_data' to include 'land area' (in sq. miles), 'water area' (in sq. miles), and latitude and longitude for each zip code, merged the dataframe 'zipcode_pop' (US Population Density by Zip Code, City, and State Dataset) onto 'realtor_data' to include 'population' and 'density' for each zip code, and merged the 'crime_data' dataframe (Offenses Known to Law Enforcement Dataset) onto 'realtor_data' to include crime statistics for each 'city_state' (a combinationation of city and state as a temp identifier), and merged the dataframe 'zipcode_pop' onto 'realtor_data' to include 'population' and 'density' for each zip code.

We then got rid of any additional columns from the merges that were unnecessary and filtered 'population' to only keep rows where the city has a population above 0. We then had to impute missing values again due to the merges, so we did this with the median for each ('city', 'state') group for the 'land_area', 'water_area', 'latitude', and 'longitude' features, and dropped the remaining rows where 'latitude' and 'longitude' were missing, as these corresponded to non-existent areas, such as out of county, louisiana or out of county, california, or remote locations such as Naukati Bay, Alaska, or Kasaan, Alaska. We then merged the 'uni_data' dataset (Institutional Characteristics Dataset) onto 'realtor_data' by grouping by zip code and computing the nearest university once for each unique zip code using the haversine distance formula:

$$d = 2r \arcsin\left(\sqrt{\sin^2(\frac{\phi_2 - \phi_1}{2}) + \cos(\phi_1)\cos(\phi_2)\sin^2(\frac{\lambda_2 - \lambda_1}{2})}\right)$$

and creating new features of characteristics of the nearest university to that zip code.

We also created two new features: 'median_zip_price', which represented the median price of homes in the zip code (helped compare listing prices to the market norm), and 'inventory_count', which represented the number of homes available for sale (low inventory $\rightarrow$ higher prices) by zip code. To deal with the remaining missing values, we decided to calculate the nearest zip code for each unique zip code using the haversine distance formula again and creating the features 'nearest_zip' and 'nearest_zip_distance' (sq miles).

[6] Property Taxes by State

We then also created new features 'total_crime' that sums up total 'violent_crime' numbers and 'property_crime' numbers for a given zip code to calculate 'crime_rate' which is based on the formula:

$$\frac{\#\text{ of total crimes reported}}{\text{population}} \times 1,000$$

meaning crime rate per 1,000 inhabitants. Our last step was merging the dataframe 'property_taxes' (Property Taxes by State Dataset) to include property taxes for each state. We also created a new feature 'house_category', which represents the type of home a property is, e.g., tiny home, investment property, family home, ranch home, luxury home, luxury estate, and other. This was done through geodemographic segmentation, which selected relevant features, enabling nuanced, region-specific property classification. The derivation came from zip code-level aggregations to classify homes based on local real estate characteristics, e.g., price, house size, crime rates, property taxes, acreage, etc. Properties labeled Luxury Estates typically exceeded local norms in price, size, and acreage. Properties were labeled Luxury Homes if they met the high-price and large-size criteria, but not necessarily acreage. Properties labeled Ranch Homes depended on having substantial land size relative to local norms. Other categories included Family Homes, which were defined by mid-range prices, adequate bedrooms, and bathrooms; Tiny Homes were identified by having a small size and being affordable; and Investment Properties were characterized by high crime, elevated property taxes, or unusually small sizes.

1.2.2 Basic Statistics

After looking at the summary statistics of the numeric features, there were some extreme outliers that had to be dealt with. We saw that max 'price' is 1×10^9, which is $1 billion, however, this is not realistic because, as of February 2024, the most expensive home in the United States is Gordon Pointe in Naples, Florida, which is listed for $295 million dollars[2][7]. Therefore, we filtered

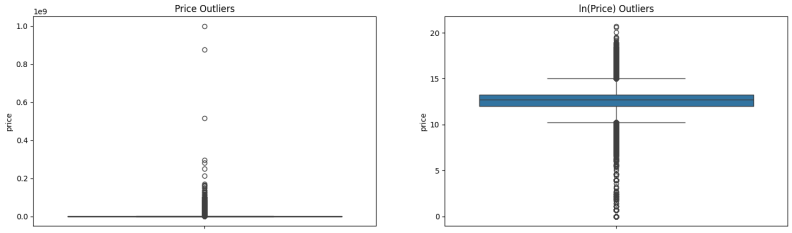

Figure 1: Box plots of Price (pre- and -post log transform)

'realtor_data' to not include the outliers in 'price' that are greater than 295 million dollars.[2] We can also see that the min is 0.0, meaning the home was free; however, this is not realistic, and to avoid data entry errors, the lowest threshold was set at $10,000. We also decided to log transform[8] 'price' because the home prices were *heavily* right-skewed (See Figure 1). We did this same process for many other numeric features that had extreme outliers, e.g., 'bed', 'bath', and 'house_size'. For the remaining columns that had relatively realistic outliers, we winsorized them to limit any extreme values by capping them at certain thresholds (we did at a 0.01% threshold level). After attempting to handle the extreme outliers, we ended up logarithmizing all of the numeric features that had a skew score above 1, and that weren't 'bed', 'bath', 'property_tax', 'crime_rate', 'nearest_zip_distance', and 'total_crime'.

1.2.3 Findings in EDA

After logarithmizing our target feature 'price', we can see that the distribution below (See Figure 2) has gotten closer to resembling a Gaussian distribution compared to the box plot (Figure 1), where we can see that the box plot of price pre-transformation is *not* normal.

[7] Most expensive home in the US is on the market

[8] By the natural log, or ln.

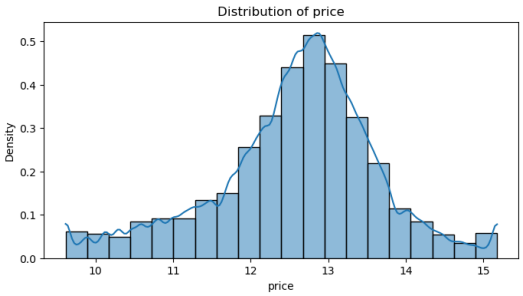

Figure 2: Histogram of (log) Price

From the correlation heatmap below (See Figure 3), we can see that a majority of the numeric features are not very correlated; however, there are a few hotspots that are highly correlated, such as 'population', 'density', 'violent_crime', and 'property_crime'.

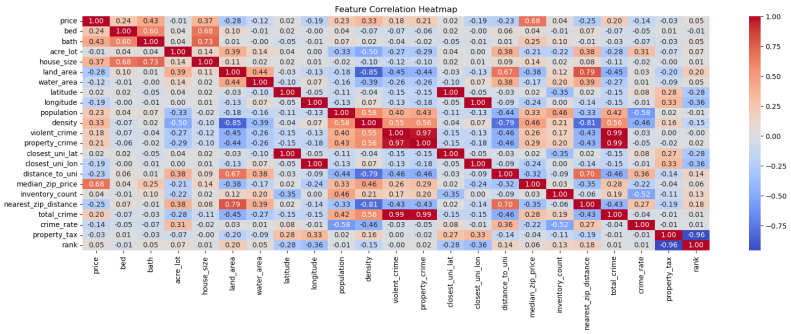

Figure 3: Correlation Heatmap

From the count plot below (See Figure 4), we can see that there are very few properties that are of the status 'ready to build', meaning these homes are to be constructed, and that around $\frac{2}{3}$ of the homes are for sale.

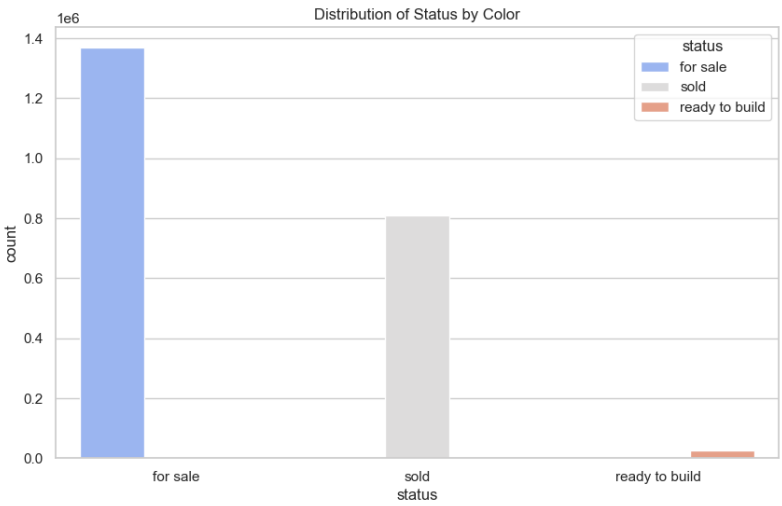

Figure 4: Distribution of Status

From the bar chart below (See Figure 5), we can see that Utah, Wyoming, and Colorado all have the highest average house size, meaning that there most likely is some correlation with the location of the state and average house size, as these 3 states are all in the Midwest.

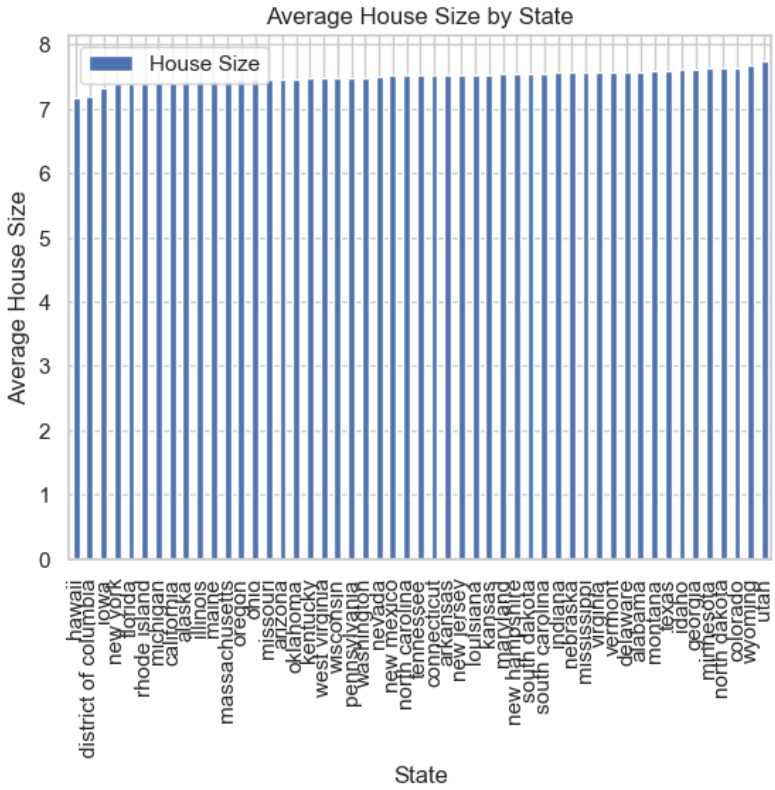

Figure 5: Average House Size by State

From the bar chart below (See Figure 6), we can see that a majority of the home prices fall within the $[12.674, 12.952]$ bin. Since these are log-transformed prices, this would equate to a bin of around \$319336.28 to \$421679.15.

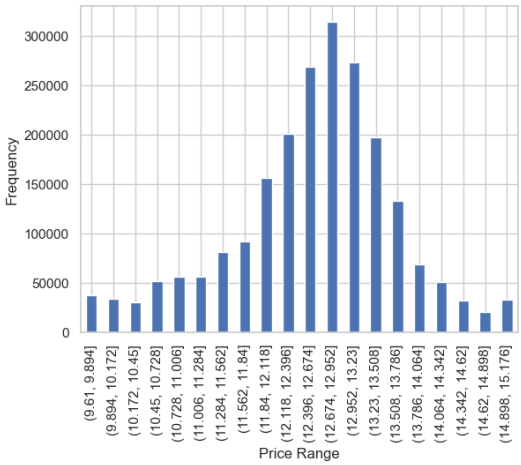

Figure 6: Price Range Frequency

From the bar chart below (See Figure 7), we can see that the states with the highest average price are DC, California, Utah, and Hawaii. This makes sense since DC is a city, and prices of an urban, dense population are typically higher. California home prices are notoriously higher because of hot spots such as San Francisco, Los Angeles, and San Diego. Utah has a high average home price because of recent high demand and limited housing supply. Hawaii having a high average home price makes sense due to scenery and high demand, since that state is a "beautiful" tropical island.

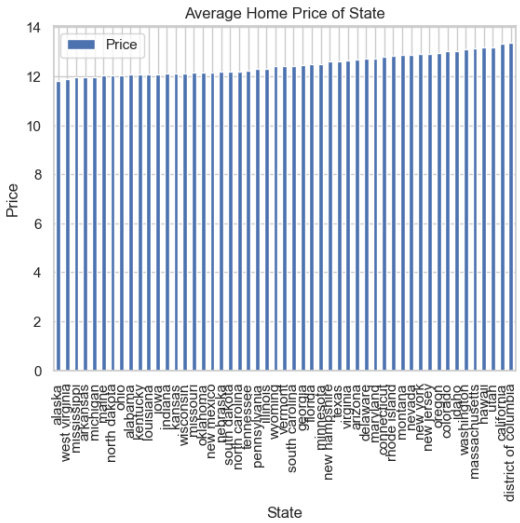

Figure 7: Average Price of State

Looking at the bar chart below (See Figure 8), we can see that it was expected that Luxury Estates command the highest prices, as they include expansive properties in prime locations. Ranch Homes appear to have a broad price distribution, which could be due to differences in land value across regions. Investment Properties show high variation, suggesting the category includes both affordable rental units and high-value investment assets. The Other category remains inconsistent, indicating that it might contain unclassified properties such as mobile homes or mixed-use spaces.

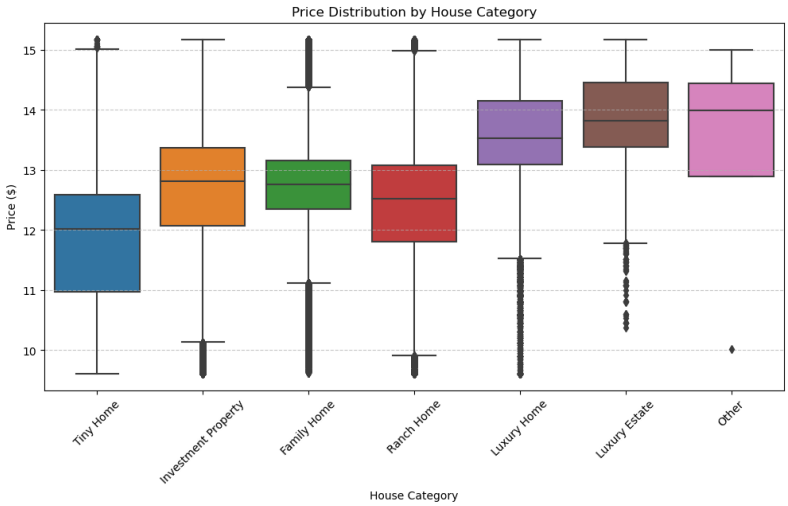

Figure 8: Price Distribution by House Category

We can see in the box plot below (See Figure 9) that Family Homes have larger median sizes than Tiny Homes, aligning with expectations, as they are designed for larger households. The wide range

in Ranch Home sizes suggests that some may be large, multi-acre properties, while others could be smaller homes with significant land. Investment Properties showing variation in size suggest a mix of rental units, duplexes, and larger income-generating properties. The relatively small size distribution of Tiny Homes reflects a niche market targeting affordability and compact living.

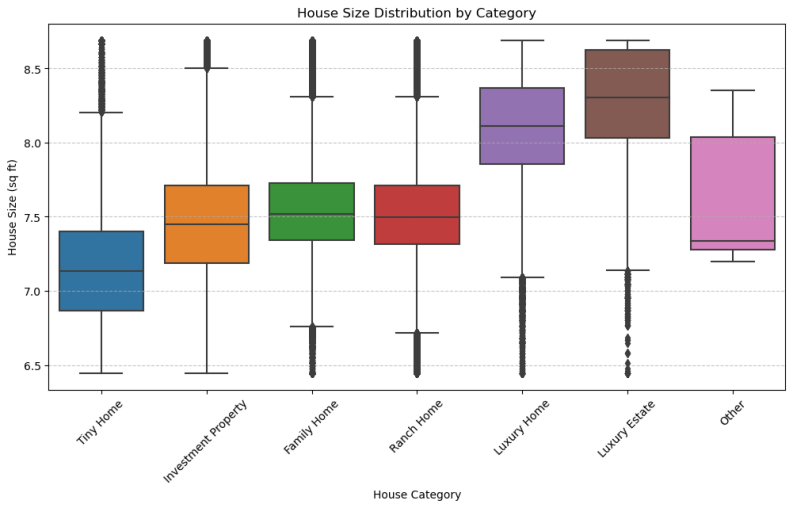

Figure 9: House Size Distribution by Category

We can see in the box plot below (See Figure 10) that Ranch Homes have large acre sizes, but mid-tier prices align with expectations, as these homes prioritize land over house size. Luxury Estates, which have both large acre sizes and high prices, appear correctly classified. The presence of large-acre properties in the Other category may indicate incomplete classifications or edge cases. The relatively small land size of Tiny Homes and Investment Properties is expected, as these properties are typically located in dense urban areas or are compact rental units.

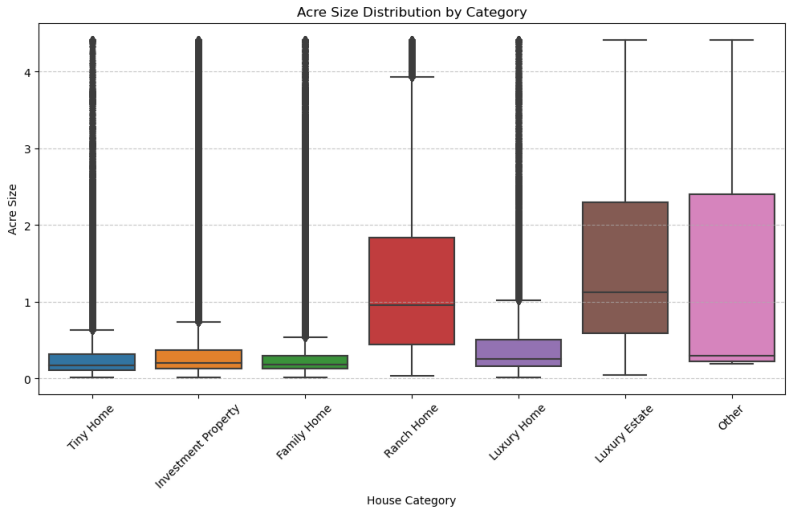

Figure 10: Acre Size Distribution by Category

2 Predictive Task

2.1 Identify a Predictive Task

Our objective is to predict *house prices* using a comprehensive array of features, including factors such as zip code, house size, and the number of bathrooms, among others. To achieve this, we aim to develop a regression model trained on an extensive dataset comprising millions of observations from across the United States. In constructing our final predictive model, we will evaluate its performance against six baseline models, iteratively improving upon them to create a more accurate and robust regression model.

2.2 Evaluation

The primary metric used to compare our baseline models and the final model is *Root Mean Squared Error*, or *RMSE*. RMSE is a commonly used evaluation metric for regression problems, providing a clear indication of model performance. It calculates the square root of the average squared differences between predicted and actual values, making it highly sensitive to large errors. This sensitivity is particularly beneficial when we want to penalize significant prediction errors more heavily.

2.3 Preprocessing Features

For our baseline models, we decided to start with the seven numerical features most correlated with price: *median_zip_price, house_size, acre_lot, bath, latitude, longitude, and crime_rate*. We used these exact features for all baseline models to standardize the comparison and make it easier to observe future improvements in terms of adding or subtracting features in the creation of our final model. We also split our dataset into training and test sets to properly evaluate our baseline models on unseen data.

2.4 Baseline Models

The first baseline model is a basic *linear regression* model. The main purpose of this model is to analyze and quantify the linear effect of each of the seven features most correlated with price. While a simple linear regression model, like this one, is not likely to be the best performing model, it is an important setting stone for more complex models with more features. Additionally, it is the model least likely to overfit.

Linear regression can help us determine the importance of our features through their weights, but sometimes these weights can be detrimental when it comes to generalizing and lead to higher variance. To prevent overfitting, we can introduce penalties in the regression and reduce the importance of the weights by shrinking them towards 0. Two techniques to achieve this are *Ridge and Lasso* regression. Therefore, we also created two more regression models as a baseline point of reference.

One of the key disadvantages of linear regression is its failure to capture nonlinear relationships, thus producing biased estimates. Because of this, we also looked at a *Random Forest Regression* model. A random forest model can effectively capture valuable trends and relationships through the use of ensemble learning methods, which train several models through a series of decision trees on random samples of the training set and return an averaged result in which the trees "vote" for the best model. However, one disadvantage is that it can lead to overfitting, so we limited max_depth to counteract it. Random forest regression can provide additional input on what features accurately predict price non-linearly.

For tabular data, *XGBoost* and *LightGBM* are known to be some of the best-performing models. Like with random forest regression, both of these algorithms use ensemble learning methods, with the addition of gradient boosting frameworks that iteratively improve predictions made by individual decision trees (or "weak learners"), with additional hyperparameters like regularization. In a way, both of these models combine previous simpler models into a faster and more accurate single model. XGBoost tends to perform better overall, but LightGBM is more memory efficient and tends to be faster for large datasets like this one.

2.5 Baseline Results

The linear regression model exhibited the poorest performance, with a validation RMSE of 0.71. We must also note that because our target variable, price, is on the log scale, an RMSE of 0.71 is *also* measured on the log scale[9]. Both the ridge and lasso regression models also recorded validation RMSEs of about 0.71, indicating that any potential improvements over the linear regression model were little to none. The random forest regression model performed slightly better, achieving a validation RMSE of 0.64. As expected, the most accurate models were XGBoost, which delivered the lowest validation RMSE of 0.57, and LightGBM, with a validation RMSE of 0.62.

3 Model

3.1 Price Prediction Model

The final model selected for the predictive task was the *Light Gradient Boosting Machine, or LightGBM*. Since our final dataset has over two million entries, LightGBM was chosen due to its superior balance between predictive accuracy and computational efficiency, making it well-suited for handling such a large dataset with numerous features. Another reason we chose LightGBM is its ability to handle categorical variables effectively and mitigate overfitting through built-in regularization techniques.

3.2 Data Preprocessing

Instead of just using the seven most correlated numerical features, like our baseline models, we decided to use all the available features, as well as engineered ones. Thus, we needed to preprocess our features before training the final model.

The prev_sold_date column, which is a datetime feature, was decomposed into three separate numerical features:

- prev_sold_year (year of the previous sale)
- prev_sold_month (month of the previous sale)
- prev_sold_day (day of the previous sale)

The original prev_sold_date column were then removed, as it had been replaced by more useful numerical representations.

To make categorial features useful for the model, we used *LabelEncoder* from *sklearn.preprocessing* to transform categorical values into numerical values[10].

3.3 Model Optimization

To make sure that our final model performed better than all of the baseline models, while avoiding overfitting, we optimized it through hyperparameter tuning.

Given that we have a large dataset, we decided to use *RandomSearchCV* from *sklearn.model_selection* over *GridSearchCV* to find the best combination of hyperparameters for our model for computational reasons.

The parameters that we ended up using for the final model were:

- **objective**: regression
- **metric**: rmse
- **boosting_type**: gbdt
- **num_leaves**: 100

[9]For interpretation, a log-scale RMSE reflects multiplicative rather than additive error. An RMSE of 0.71 corresponds to predictions typically being off by approximately a factor of two on the original price scale, meaning predicted home prices are often about half or double the true value.

[10]E.g, if a column contains ['Apartment', 'House', 'Condo'], LabelEncoder would convert it to [0, 1, 2].

- **max_depth**: 20
- **learning_rate**: 0.1
- **lambda_l1**: 1.0
- **lambda_l2**: 1.0
- **feature_fraction**: 1
- **bagging_fraction**: 0.8

3.4 Mapping Valuation

We computed the difference between the model-predicted expected price and the listed price in log space, defined as

$$d = \log \hat{P} - \log P.$$

This quantity represents the logarithm of the ratio between the expected and listed prices. Properties with differences within ± 1 standard deviation of the overall distribution of d were classified as *properly valued*. Properties with $d < -1\sigma$ were classified as *overvalued*, indicating that the listed price substantially exceeded the model's expected price, while properties with $d > 1\sigma$ were classified as *undervalued*, suggesting that the listed price was meaningfully lower than expected.

Using this valuation framework, approximately 79% of homes were classified as properly valued. In contrast, about 10.5% were identified as undervalued, potentially representing attractive investment opportunities. This distribution indicates that while most homes are priced close to their model-implied value, a nontrivial fraction exhibits substantial deviations, highlighting the importance of accurate price estimation in real estate decision-making.

4 Literature

Implementing GIS in Real Estate Price Prediction and Mass Valuation: The Case Study of Nicosia District. By Charambolos Yiorkas and Thomas Dimopoulos
https://www.researchgate.net/publication
Yiorkas and Dimopoulos wanted to produce better real estate predictions for a small dataset of 1341 properties in Cyprus. The main purpose of this report was to prove that geospatial features such as proximity to schools or universities had a significant effect on property prices. Price predictions for properties in Cyprus were usually done by only looking at the physical properties of the house, such as its size, age, and number of bedrooms and bathrooms. The two main models that were performed in this research were Ordinary Least Squares Regression (as a baseline model) and Geographically Weighted Regression (as the final model). This paper served as inspiration for us to include features that were outside of our original dataset and related to the geography and environment of the property. The main evaluation metric used was R-squared rather than RMSE, and the Geographically Weighted Regression model proved to be far more accurate, with an R-squared value of 0.79 versus 0.55 for the OLS model, thus proving that in some instances, geographical features can be a very useful addition to a property price prediction model.

Advanced Machine Learning Algorithms for House Price Prediction: Case Study in Kuala Lumpur. By Shuzlina Abdul Rahman, Sofanita Mutalib, Ismail Ibrahim, Nor Hamizah Zulkifley
https://www.researchgate.net/publication
This project was the most similar to ours that we could find in terms of data preprocessing, feature selection, and the models that were implemented. It is based on a dataset of 53883 entries representing individual properties in Kuala Lumpur, and the four models that were implemented were Multiple Linear Regression, Ridge Regression, LightGBM, and XGBoost, which turned out to be the best performing model. The main difference between our project and this one is the missing value imputations. Rahman, Mutalib, Ibrahim, and Zulkifley opted to simply drop all rows with any missing values, which significantly reduced the size of the dataset to 31899 entries, whereas we followed a multi-step implementation process that attempted to impute missing values based on local group medians (such as zip code), or global medians if not possible. Price was also log transformed due to extreme skewness, showcasing the many similarities between our project and

this one, even while taking into account the substantial geographical differences (Kuala Lumpur, a city, vs the United States, a large country). Overall, this project shared many commonalities with our regression component of the project, but it doesn't go into the value analysis (whether the home is undervalued, overvalued, or neither).

5 Results

5.1 Model Comparison

Our analysis compared several predictive models for house price estimation. The baseline linear regression model (OLS) produced an RMSE of 0.71, indicating limited capabilities with respect to capturing non-linear relationships. The regularizing linear regression models (Ridge and Lasso) resulted in minimal improvements, if any, with RMSEs close to 0.71, suggesting that simply penalizing coefficients did not enhance predictive accuracy. In contrast, the ensemble models performed significantly better; the Random Forest model achieved an RMSE of approximately 0.64, the XGBoost model attained the lowest RMSE at around 0.57, and the LightGBM delivered an RMSE near 0.62. Taking the improved computational efficiency on large datasets into account, we decided to use LightGBM as the final model, hypertuning it for a final RMSE of approximately 0.4. Implying that predictions are typically within a multiplicative factor of about 1.5 of the true home price on the original scale. While individual predictions may deviate substantially, the model still captured the relative price structure well across the dataset.

5.2 Discussion

Our results underscore the importance of both model choice and careful interpretation of predictive accuracy in housing price modeling. Incorporating external geographic and environmental features consistently improved performance across all model classes, confirming that housing prices are shaped by spatial and contextual factors beyond basic structural attributes. Explicitly encoding this information reduced unexplained variation that simpler feature sets failed to capture.

As expected, baseline linear models (OLS, Ridge, and Lasso) underperformed relative to the ensemble-based models. While linear models offer interpretability, their restrictive functional form limits their ability to capture the complex, non-linear interactions inherent in real estate markets. In contrast, the tree-based ensemble methods adapted more effectively to these relationships and achieved substantially lower error.

Among all models evaluated, LightGBM delivered the strongest performance, achieving a final RMSE of approximately 0.4 on the log-transformed prices. Its combination of accuracy and computational efficiency made it the preferred final model, though this performance comes at the cost of interpretability. As a largely black-box approach, LightGBM offers limited direct insight into how individual features influence predictions. Future work should therefore focus on post hoc interpretability techniques, such as SHAP values or partial dependence plots, to better understand how the model captures proportional price effects in log space.

Interpreting model performance in the log-transformed target space is critical. Because predictions are evaluated multiplicatively rather than additively, an RMSE of 0.4 implies that predicted prices are typically within a factor of roughly 1.5 of the true value. While this can translate into sizable absolute dollar errors for individual properties, it reflects reasonable relative accuracy given the inherent volatility of housing markets. Consequently, the model is better suited for identifying broad pricing patterns, comparisons, and rankings than for producing precise point estimates for individual homes. This suggests that future evaluations may benefit from task-specific metrics, such as ranking accuracy, segment-level error, or probabilistic uncertainty estimates, rather than relying solely on global RMSE.

Our approach to missing data also contributed to model robustness. By using localized median imputation instead of listwise deletion, we preserved a larger and more representative sample, reducing potential bias and improving stability. Building on this foundation, future work could explore richer feature representations, including higher-resolution neighborhood characteristics, temporal market indicators, or hierarchical and spatial modeling frameworks that explicitly account for the nested

structure of housing markets. Such extensions may further reduce log-scale error while improving generalization to new geographic areas.

References

Datasets:

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- https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.html
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- https://www.fourfront.us/data/datasets/us-population-density/
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